ITC327W: Work Integrated Learning

Phase 1 Portfolio

Project Title: Big Brother Student Accommodation Management System

Stakeholder: Mr Ngubane

Organisation: Big Brother Student Accommodations

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Submission Date: 04/09/2026

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1. INTRODUCTION

1.1 Purpose of the Project

Big Brother Student Accommodation currently experiences difficulties in advertising accommodation, sharing accommodation information, managing student applications, and maintaining accommodation records. Most processes are handled manually and rely heavily on word-of-mouth communication, WhatsApp groups, and referrals. The purpose of this project is to develop an integrated accommodation management solution that improves access to accommodation information, simplifies applications and bookings, and assists management in monitoring accommodation operations.

1.2 Proposed Solution

The proposed solution consists of:

- Flutter Mobile Application
- ASP.NET Web Application
- Shared Supabase Backend

The two applications will share information through a centralized Supabase database, ensuring data consistency between students and accommodation management.

2. STAKEHOLDER PROFILE

Name: Mr Ngubane

Role: Owner/Property Manager

Responsibility: Full Operational management, student sourcing, bookings, payments, data management

Current Admin Time: Manual Systems with no proper management tools in place

2.1 Organisation Information

ITEM	DETAILS
Business Name	Big Brother Student Accomodations
Stakeholder	Mr Ngubane
Business Type	Student Accommodation Provider
Location	Universtas, Bloemfontein
Primary Users	Management
Secondary Users	Student and Prospective Tenants
Contact	068 518 4009

Information obtained from stakeholder interviews indicates that the organisation provides NSFAS-accredited accommodation mainly for UFS and CUT students. Shuttle services are also available. The organisation relies heavily on referrals, WhatsApp groups, current students, and taxi association for advertising.

3. CURRENT PROCESS ANALYSIS

3.1 Current Process Description

The current accommodation process operates as follows:

1. Big Brother Student Accommodations advertises its accommodation through word of mouth, WhatsApp groups, current students living there, group administrators of the shuttle service taking students to their hometowns when schools close.

2. Prospective students make enquiries through these informal communication channels.

3. Students are encouraged to visit the accommodation in person for a viewing.

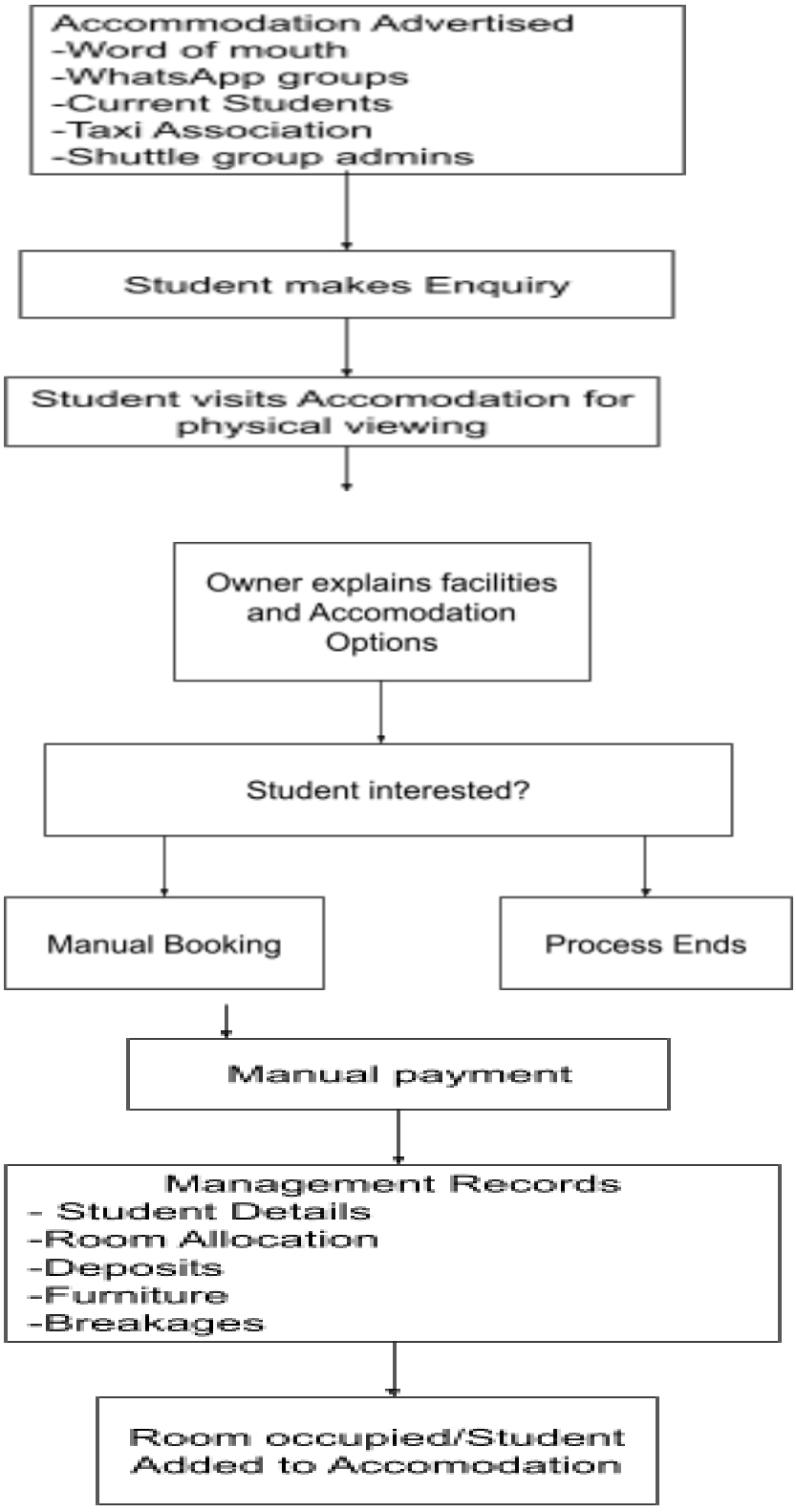
4. During the viewing, the owner explains the available facilities and accommodation arrangements.

5. The student decides whether the accommodation is suitable.

6. Bookings and payments are currently handled manually because the organisation does not have a formal management system.

7. Management keeps information such as available units, room furniture, student details, deposits, and breakages manually.

3.2 Current Process Flow Diagram



3.3 People Involved

- **Accommodation owner and management**: Advertises the accommodation, explains the facilities, manages student information, and controls applications and bookings.
- **Current students living there**: Help advertise the accommodation to prospective tenants.
- **Taxi association shuttle organisers**: Help promote the accommodation to students travelling to the university at the start of each academic year.
- **Shuttle group administrators**: Share accommodation information with potential tenants.
- **Prospective tenants**: View the accommodation, submit their details, apply, and make booking decisions.

4. PROBLEM ANALYSIS

4.1 Problem Statement

Big Brother Student Accommodations currently relies on informal advertising methods such as word of mouth, WhatsApp groups, and taxi association promotion. Prospective students generally need to visit the accommodation physically to view the rooms and receive information. Bookings and payments are also managed manually because the organisation does not have a central management system. This process makes it difficult for the organisation to reach enough prospective students, share accurate accommodation information, manage applications, and monitor available units. The stakeholder also reported concerns about first-year students being targeted by people falsely claiming to assist with accommodation arrangements. Management does not currently have a central system for managing student information, deposits, room occupancy, furniture, and breakages.

Big Brother Student Accommodation lacks an integrated digital system to manage its core operations, resulting in the following specific problems:

Identified Problems:

- **Difficulty sourcing students** - The business struggles to reach the intended number of students. Currently, advertising is mostly done through word of mouth,

WhatsApp groups, and taxi associations. This makes it difficult to attract enough students, especially those who are new to the university or coming from other areas. As a result, rooms often remain empty, which affects the business financially. Many students simply do not know that this accommodation exists.

- **Inaccessibility for Students** - Students cannot easily view the accommodation before deciding. The business relies on physical viewings, which are not always possible for students who are far away or do not have transport. This discourages many students from considering the accommodation because they cannot see what they are getting. It also creates a barrier for first-year students who may not yet be familiar with the area.

- **First-year students are vulnerable to Scammers -** Because there is no official online platform for accommodation applications, first-year students are often targeted by scammers who pretend to help them find accommodation. These scammers take money from desperate students and disappear, leaving them without a place to stay. This damages the reputation of legitimate accommodation providers and makes students afraid to trust anyone when looking for housing.

- **Difficulty sharing information -**Students cannot easily find information about the accommodation online

- **No proper management system -** Bookings and payments are manual, causing confusion and delays

- **Business continuity threat -** These challenges pose a real threat to the existence of the business. If students cannot find the accommodation, cannot easily apply, or do not feel safe using the current process, they will look elsewhere. The business may struggle to remain sustainable if it cannot attract and retain enough students each year**.**

- **No application platform -** Students cannot apply online or track their application status

- **Limited access to information** - Management cannot easily access or respond to student applications

4.2 Consequences

The current process may result in:

- Limited reach to prospective tenants
- Difficulty sharing accommodation information
- Delays in managing applications and bookings
- Inconsistent or incomplete records
- Difficulty tracking room availability and student occupancy
- Increased exposure of students to accommodation scams
- A possible threat to the organisation’s business continuity

4.3 Need for Improvement

The organisation needs a centralised system that makes accommodation information easier to access, allows prospective tenants to view availability and apply, and helps management control, monitor, and respond to student information. The system should also protect student information and support more effective marketing of the accommodation.

5. PROJECT AIM AND OBJECTIVES

5.1 Overall Purpose

To develop an integrated software prototype that digitises and streamlines the management operations of Big Brother Student Accommodation, improving student sourcing, application processing, and operational efficiency.

5.2 Specific Objectives

Objective	Action
1. Digital application platform	Enable students to upload particulars and apply through the system
2. Accommodation viewing	Allow students to view accommodations, check availability, and make bookings
3. Centralised management	Enable management to control, monitor, and manage data
4. Information sharing	Allow easy sharing of business information to prospective tenants
5. Security	Secure student information and ensure it is not accessible by unauthorised persons
6. Application tracking	Allow students to check status updates accordingly
7. Data analysis	Enable management to access, analyse, and respond to student information
8. Student self-service	Allow students to view, add, and change their information
9. Business visibility	Allow students to understand vision, values, and mission of the organisation
10. Payment system	Implement payment functionality
11. Feedback mechanism	Collect reviews, comments, and areas of improvement from students
12. Unit management	Track units available, furniture, occupant details, deposits, and breakages

6. PROJECT SCOPE

6.1 Deliverables

- Flutter mobile application for the management and students
- ASP.NET web application for management
- Supabase shared backend and database
- Integrated solution where both applications access the same data
- Technical documentation
- User guides where applicable

6.2 Acceptance Criteria

- Management can view and manage student applications
- Management can control, monitor, and manage data
- Management can access, analyse, and respond to student information
- Management can view accommodation listings and occupant details
- Students can upload particulars and apply on the system
- Students can view accommodations, check availability, and make bookings
- Students can view, add, and change their information (with management approval)
- Students can submit reviews, comments, and recommendations
- Student information is secured and not accessible by unauthorised persons
- All information is consistent across mobile and web applications
- System is functional for demonstration purposes

6.3 Exclusions

- Full payment processing integration (e.g., automated EFT reconciliation)
- Advanced reporting and analytics dashboards
- Integration with external systems (e.g., banking APIs, municipal systems)
- Full production deployment and hosting
- Professional security audit and penetration testing
- Mobile app store publication
- Multi-property support (limited Big Brother Student Accommodation)

6.4 Project Constraints

- **Time:** Development must be completed within one academic semester
- **Budget:** Limited budget (student project)
- **Technology:** Technology stack is predetermined: Flutter, ASP.NET, Supabase
- **Skills:** Team comprises IT students possessing Flutter, Dart, ASP.NET, Github and Supabase skills necessary for development of the system.

6.5 Project Assumptions

- Stakeholder will be available for requirements gathering, feedback, and evaluation
- Student users will be accessible for feedback and testing
- Supabase will provide sufficient functionality for the project scope
- The approved prototype does not need to be production-ready
- All development work will be managed through GitHub
- Microsoft Project will be used for project scheduling

7. STAKEHOLDER NEEDS

- Students should be able to upload their particulars and apply on the system
- Students should be able to view accommodations, check availability, apply, and make bookings
- Students should be able to check status updates
- Management should be able to control, monitor, and manage data
- Management should be able to access, analyse, and respond to information
- Prospective tenants should be able to use the application
- Management should be able to use the application
- Students should be able to view the entire processes and business model
- Students should understand the vision, values, and mission of the organisation
- Student information should be secured and not accessible by unauthorised persons
- Easy access to information and application for accommodation
- Payment system should be included
- Students should be able to view, add, and change their information (with management approval)
- Management should be able to view accommodation listings and occupant details
- System should provide summary of students visiting page and viewing accommodation
- System should collect reviews, comments, and areas of improvement from students
- System should track units available, furniture, occupant details, deposits, and breakages

8. FUNCTIONAL REQUIREMENTS

1. FR1 System shall allow students to create accounts and upload their particulars
2. FR2 System shall allow students to submit accommodation applications
3. FR3 System shall allow students to view available accommodations
4. FR4 System shall allow students to check availability and make bookings
5. FR5 System shall allow students to check application status updates
6. FR6 System shall allow management to control, monitor, and manage data
7. FR7 System shall allow management to access, analyse, and respond to student information
8. FR8 System shall display the entire application process and business model
9. FR9 System shall display the organisation's vision, values, and mission
10. FR10 System shall authenticate users (management and student roles)
11. FR11 System shall secure student information
12. FR12 System shall provide easy access to information and applications
13. FR13 System shall include payment functionality (Medium)
14. FR14 System shall allow students to view, add, and change their information
15. FR15 Management shall approve changes made by students
16. FR16 allow management to view accommodation listings and occupant details
17. FR17 provides student accommodation viewing summaries.
18. FR18 System shall collect student feedback and improvement areas.
19. FR19 Track availability, furnishings, occupants, deposits, and damages.

9. NON-FUNCTIONAL REQUIREMENTS

1. NFR1 **Usability:** Mobile application must be responsive for student use
2. NFR2 **Usability:** Web application must be accessible via standard web browsers
3. NFR3 **Usability:** System should have intuitive user interfaces requiring minimal training
4. NFR4 **Security:** System must protect student personal information
5. NFR5 **Security:** Authentication must separate management and student access levels
6. NFR6 **Availability:** Supabase database must be accessible to both Flutter and ASP.NET applications
7. NFR7 **Reliability:** System must maintain data integrity across all platforms
8. NFR8 **Maintainability:** Source code must be organised and documented

COULD HAVE REQUIREMENTS

ID	Requirement	Description	Reason
C1	Advanced Online Payment	Students could complete additional payment functions online.	Could improve convenience but requires additional implementation considerations.
C2	Reviews and Ratings	Students could review and rate the accommodation.	Useful for future marketing and engagement.
C3	Advanced Reports	Management could generate reports on applications, bookings and occupancy.	Useful for future business analysis.

C4	Virtual Accommodation Tour	Students could view the virtual tours of accommodation facilities.	Could further reduce dependence on physical viewing.
C5	Favorites	Students could save preferred accommodation listings.	Convenience feature that is not essential to the core system.

10. INTEGRATION REQUIREMENTS

How Flutter, ASP.NET and Supabase Will Operate as One Integrated Solution

The proposed solution for Big Brother Student Accommodation will consist of three main components working together seamlessly: a Flutter mobile application, an ASP.NET web application, and a Supabase backend. These components will operate as one integrated system where all data is shared and synchronised in real-time.

Supabase Backend (Shared Database & Authentication):

Supabase will serve as the central backbone of the system, providing database, authentication, real-time subscriptions, row-level security, and API endpoints for both Flutter and ASP.NET applications to interact with**.**

Flutter Mobile Application (Students & Management):

The Flutter mobile application will allow students to view accommodations, check availability, apply, make bookings, track application status, upload particulars, and submit reviews. Management will be able to view and manage applications, control data, monitor bookings, access analytics, manage units, and respond to students. Flutter will use the Supabase Dart client to connect directly to the Supabase backend, with all data retrieved and displayed in real-time**.**

ASP.NET Web Application (Management):

The ASP.NET web application will provide management with a dashboard, student management, unit management, payment tracking, reporting, and data management capabilities.ASP.NET will connect to Supabase using the Supabase C# client or REST API, accessing the same database tables as the Flutter application.

Integration Flow:

When a student applies through the Flutter app, the data is sent to Supabase, stored in the database, and a real-time update triggers the ASP.NET dashboard so management can see the new application. When management approves or rejects the application through ASP.NET, Supabase updates the status, and a real-time update triggers Flutter so the student can see the status change**.**

Integration Requirements

Shared Database - Both Flutter and ASP.NET applications shall connect to the same Supabase database

Data Consistency - Information updated on one platform must be immediately available on the other platform

Authentication - User authentication managed through Supabase Auth, enabling login from both platforms

Shared Schema - Database schema designed to support both management and student functions

API Consistency - Both applications use consistent API endpoints and data formats

Real-time Updates - Supabase real-time subscriptions keep both applications synchronised where appropriate

REQUIREMENTS CHANGE LOG

During Phase 1, several adjustments were made to the project requirements based on feasibility analysis and project constraints. The following changes were documented and approved to ensure the project remains realistic and achievable within the available timeframe and resources.

Payment Processing Integration:

The original requirement for full payment-gateway integration was revised to simulated payment-status data. This change was necessary because implementing full payment processing would have introduced significant technical complexity that exceeded the project's constraints. The team determined that demonstrating payment functionality using simulated payment-status data would be more appropriate for the prototype scope, allowing the stakeholder to see how payment tracking would work without the complexities of integrating with banking systems.

Automated Contract Signing:

The requirement for automated contract signing was removed and declared out of scope. This decision was made because implementing digital contract signing would require additional legal

considerations and technical complexity that are not feasible within the academic project timeline. The team and stakeholder agreed that the prototype should focus on core functionality rather than attempting to implement features that would add unnecessary complexity.

Advanced Analytics and Document Verification:

The requirements for advanced analytics and complex document verification were removed and declared out of scope. The team recognised that building complex dashboards with sophisticated data visualisation and document verification would require extensive development effort that could not be justified for a prototype. This change ensures that the development effort is directed towards the most critical features.

Multi-Property Expansion:

The requirement for multi-property expansion was removed and declared out of scope. The team and stakeholder agreed that the prototype should focus on a single property (Big Brother Student Accommodations) to keep the project manageable within the academic timeline

REFINED PROJECT SCOPE

PROJECT SCOPE INTRODUCTION

Big Brother Student Accommodation is a student housing business located in the Universitas area of Bloemfontein, owned by Mr Ngubane. The business faces significant challenges in sourcing students, managing bookings and payments, processing applications, and ensuring the security of student information. This project aims to develop an integrated software prototype that will digitise and streamline the accommodation management processes, improve student sourcing, enhance communication, and reduce administrative overhead.

The solution will consist of a Flutter mobile application for students and an ASP.NET web application for management, supported by a shared Supabase backend to ensure information consistency across both platforms.

11. FEASIBILITY STUDY

Feasibility Study (Technical, Operational, Economic, Schedule, Legal, and Ethical feasibility studying)

The proposed student accommodation management system is feasible with minor changes, directly solving the stakeholder's core operational, marketing, and scam-prevenation issues.

Feasibility Area	What Must Be Evaluated
Technical feasibility	Highly feasible. The tech stack (Flutter for frontend, ASP.NET Core for backend APIs, Supabase for cloud database and authentication) directly supports all required capabilities, including online room viewing, application submissions, real-time availability tracking, and admin management portals.
Operational feasibility	Highly feasible. The application directly addresses critical pain points by modernizing their reliance on manual bookings and word-of-mouth marketing (points 2, 3 and 4 found under section: Stakeholder Needs). It eliminates distance barriers by providing remote viewing, mitigates scam risks targeting first years (point 5 Stakeholder Needs), and streamlines how management tracks and responds to applications (point 9, 12 and 13 of Stakeholder needs).
Economic feasibility	Feasible with scope control. Developing an open-source/low-tier backend (Supabase free tier) keeps costs near zero during the academic build phase. Budgeting focuses solely on developer time and free-tier infrastructure.
Schedule feasibility	Feasible with scope control. Building a full end to end payment gateway within a single academic semester poses schedule risks. To ensure delivery, the members should implement stimulated payment status updates for phase 1 rather than full bank/gateway integration.

Legal feasibility	Feasible. Handling prospective student personal information (IDs, proof of registration, contact details) requires strict adherence to privacy legislation, for example POPIA/GDPR. Secure data storage via Supabase with role based access control “RBAC” ensures compliant data handling.
Ethical feasibility	Highly feasible. Digitalizing the platform protects vulnerable first year students from accommodation scams (point 5: Stakeholder Needs) and creates a fair, transparent application process for all applicants regardless of their physical location.

Overall Feasibility Decision

Justification: The core system solves direct threats to the stakeholder’s business survival. However, full operational features like live automated payment gateways and complex admin analytics should be scaled back to mock features/prototypes to maintain schedule and technical feasibility within the semester timeframe.

12. RISK REGISTER

12.1 Risk Register:

	A	B	C	D	E	F	G	H	I	J	K
1	Risk ID	Risk statement	Category	Likelihood (1-3)	Impact (1-3)	Score	Level	Mitigation	Contingency	Owner	Status
6	R5	too many features (eg. online payments)	Scope	2	3	6	Medium	must,should and could have and confirm core scope with stakeholder early	Remove or postpone lower-priority	Person 3/4	Open
7	R6	Github merge conflicts may occur due to 10 members pushing the same repo	Team	3	2	6	Medium	Create branch per member and person 9 reviews pull requests	Revert to last stable main branch and re-apply changes	Person 9	Open
8	R7	A group member may not complete assigned task on time, delaying dependent activities	Team	2	2	4	Medium	Assign clear individual responsibilities and monitioring progress weekly via the progress tracker	Reassigning urgent tasks to available members and record the concern in the tracker	Person 10	Open
9	R8	Poor communication between group members may cause delays in task handover	Team	3	2	6	Medium	Schedule fixed meeting time (eg. every friday) have daily update on Whatsapp group chat and send reminders 24 hours before deadline	If member does not respond 2 members will be responsible for contact the member directly and record communication concern in progress	Person 10	Open
10	R9	Real student personal information may be accidentally uploaded to Github	Data & Security	1	3		Low	Use only fake test data, check commits before push	Immediately delete file purge history and replace with test data	Person 7	Open
11	R10	Limited testing time before the final demonstration could mean booking or approval bugs go unnoticed	Quality	2	3	6	Medium	Schedule dedicated testing time in the MS Project schedule separate from development time	Focus testing on core must-have function (booking and approval) before the demo	Person 8/9	Open
12											
13											

11.2 Top 3 Highest Risks:

	A	B	C
1	High Risk	Why Is It Important ?	What must the Group do now?
2	1. R02 - Double-booking may occur if availability check logic fails	If availability logic fails,two studenst will be booked for the same room and the stakeholder will reject the prototype	Person 3 must test the Supabase date-overlap query early using 20 fake booking and share screenshot in group
3	2. R04 Limited Supabase experience may delay backend integration between Flutter and ASP.NET	If Supabase integration fails, both Flutter mobile app and ASP.NET web app will not receive data and the whole project stops	Person 9 must complete Supabase tutorial in Week 1 and build a small test integration before full development
4	3. R08 poor communicatuion between group members may cause delays in task handover	With 10 members, late replies on Whatsapp cause next member to wait, this delays dependent activities like ASP.NET admin approval depending on Flutter booking. It also affects MS Project Schedule and cause merge conflicts.	Person 10 to schedule weekly progress meetings every Friday, on teams or physically, enforce dailly updates on Whatsapp and send reminders on deadlines 24 hours before deadline
5			
6			

13.WORK BREAKDOWN STRUCTURE (WBS)

1. Planning and Requirement Analysis

1.1 Stakeholder Identification

1.1.1 Identify stakeholder

1.1.2 Obtain stakeholder approval

1.1.3 Arrange stakeholder meeting

1.2 Stakeholder Engagement

1.2.1 Conduct stakeholder interview

1.2.2 Record interview responses

1.2.3 Collect communication evidence

1.3 Current Process Analysis

1.3.1 Analyse current accommodation process

1.3.2 Identify people involved

1.3.3 Develop current process flow diagram

1.4 Requirements Gathering

1.4.1 Gather stakeholder needs

1.4.2 Identify functional requirements

1.4.3 Identify non-functional requirements

1.4.4 Identify integration requirements

1.5 Software Requirements Specification

1.5.1 Compile SRS document

1.5.2 Review SRS

1.5.3 Finalise SRS

2. Feasibility Study

2.1 Technical Feasibility

2.2 Operational Feasibility

2.3 Economic Feasibility

2.4 Schedule Feasibility

2.5 Legal Feasibility

2.6 Ethical Feasibility

2.7 Feasibility Report Compilation

3. Risk Management

3.1 Risk Identification

3.2 Risk Analysis

3.3 Risk Register Development

3.4 Risk Mitigation Planning

4. Project Planning

4.1 Create Work Breakdown Structure

4.2 Develop Microsoft Project Schedule

4.3 Define Project Milestones

4.4 Establish Baseline Schedule

5. Project Setup

5.1 Create GitHub Repository

5.2 Add Team Members

5.3 Upload Initial Documentation

5.4 Create Supabase Project

5.5 Configure Database

5.6 Configure Authentication

6. Phase 1 Portfolio

6.1 Compile Stakeholder Analysis

6.2 Compile SRS

6.3 Compile Feasibility Study

6.4 Compile Risk Register

6.5 Gather Supporting Evidence

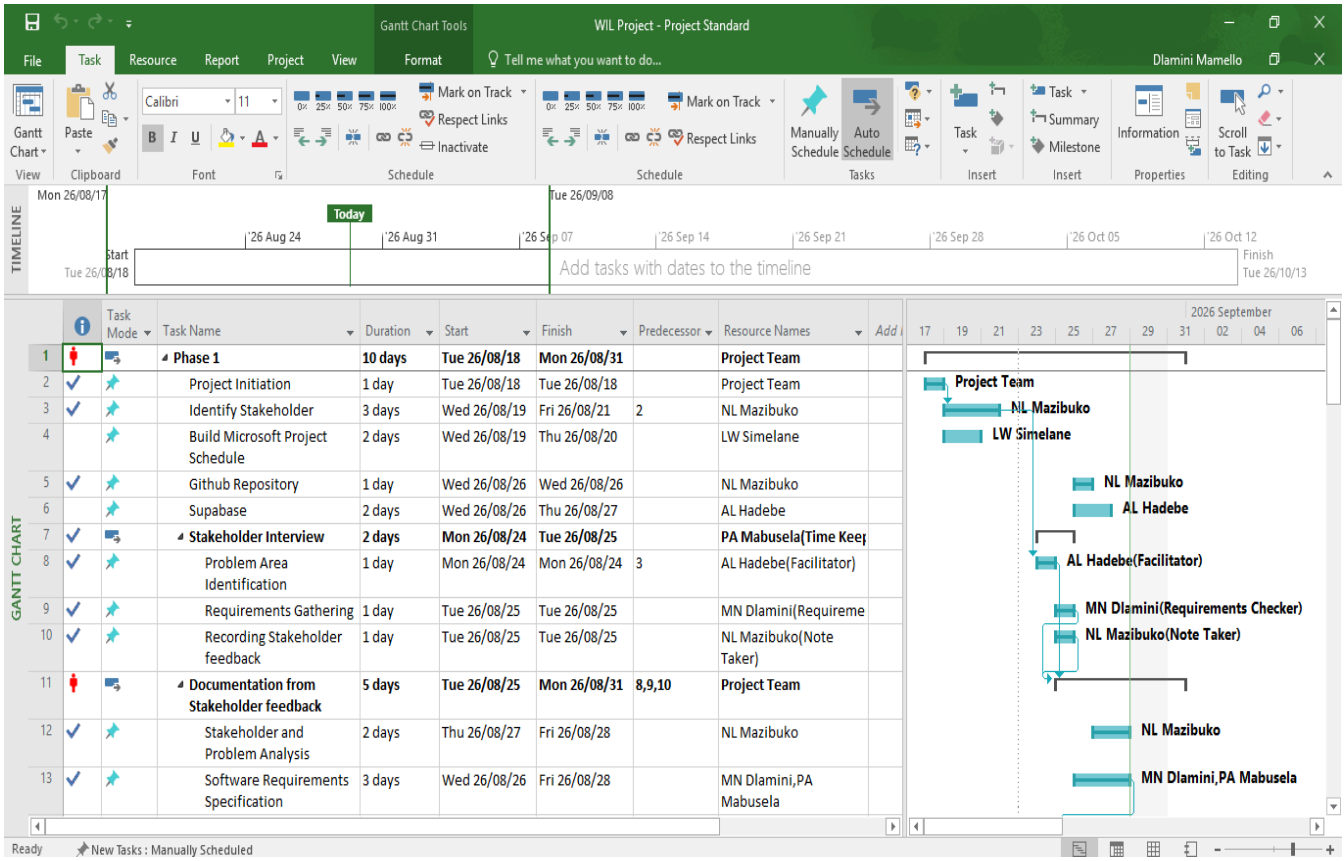
6.6 Complete Group Reflection

6.7 Final Submission

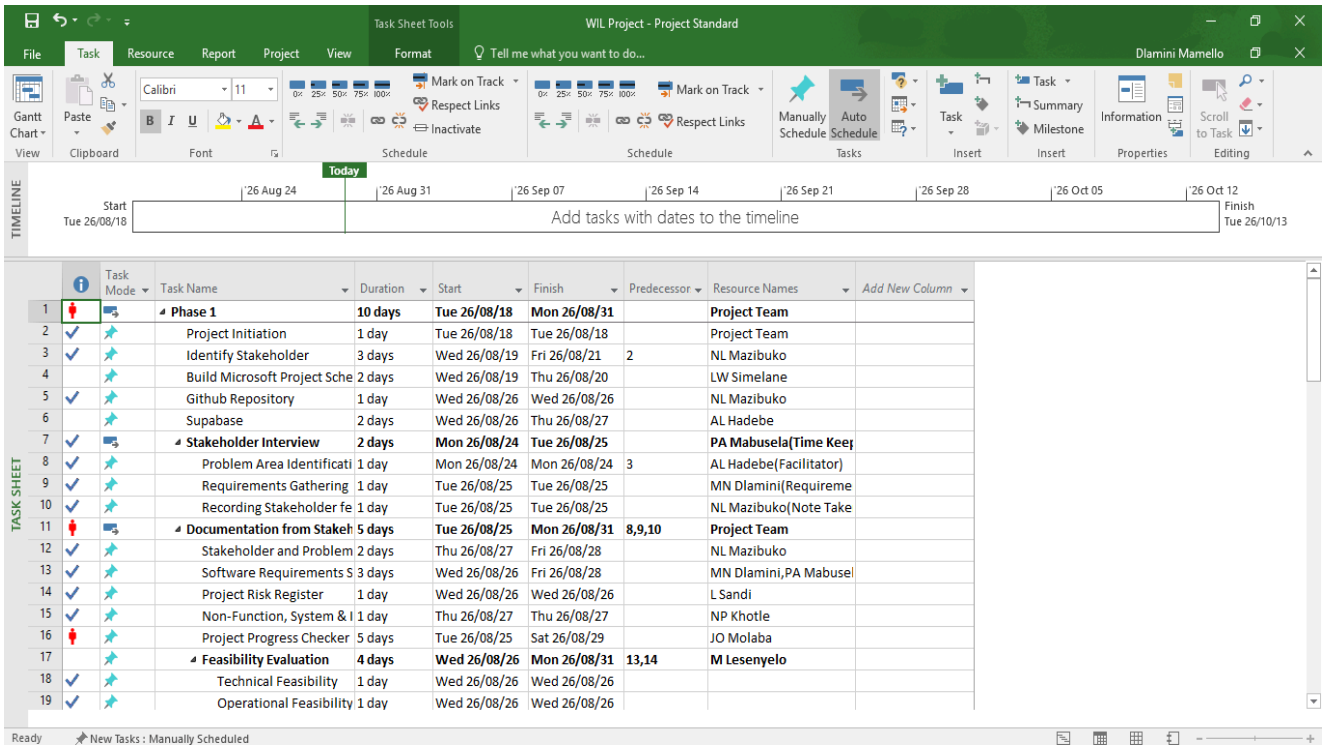
14. MICROSOFT PROJECT BASELINE

The project schedule is required to be developed and maintained using Microsoft Project to support project planning, task allocation, scheduling, tracking, and monitoring throughout the project lifecycle. The schedule will serve as the baseline against which project progress will be measured

14.1 Gantt chart:

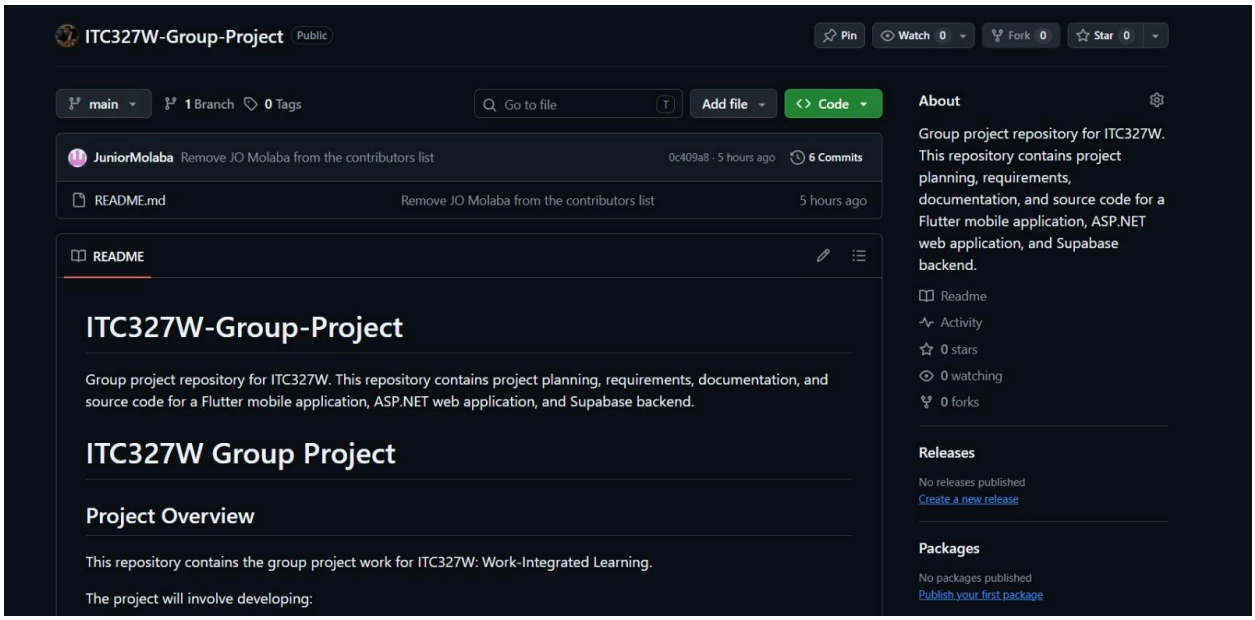


14.2 Task Sheet:



15.GITHUB REPOSITORY EVIDENCE

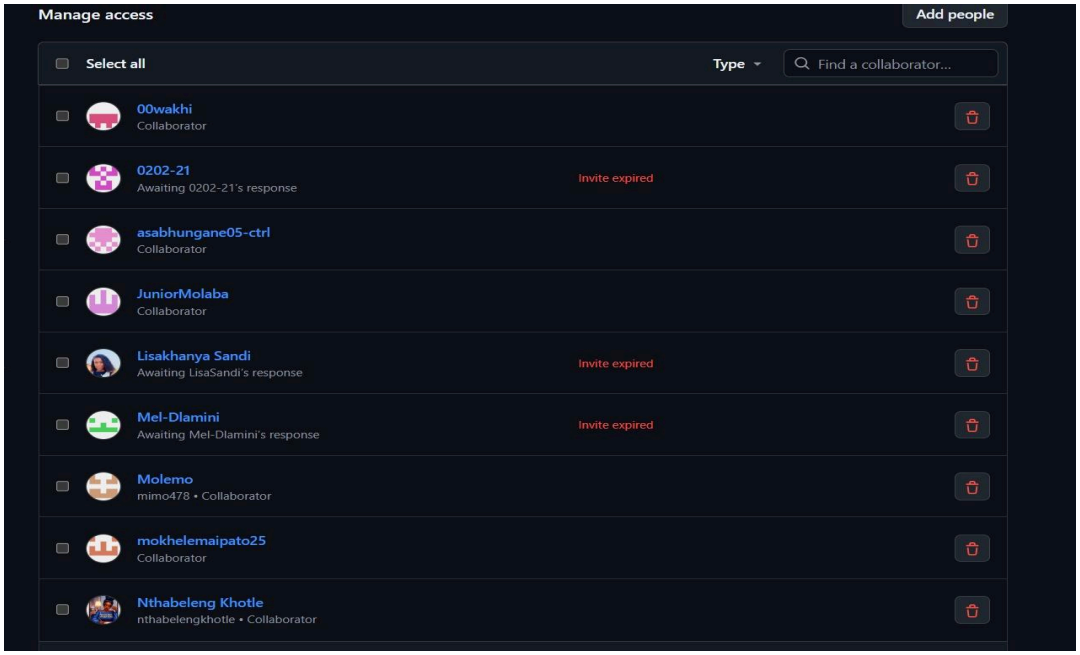
15.1 Repository Home Page:



15.2 Repository link:

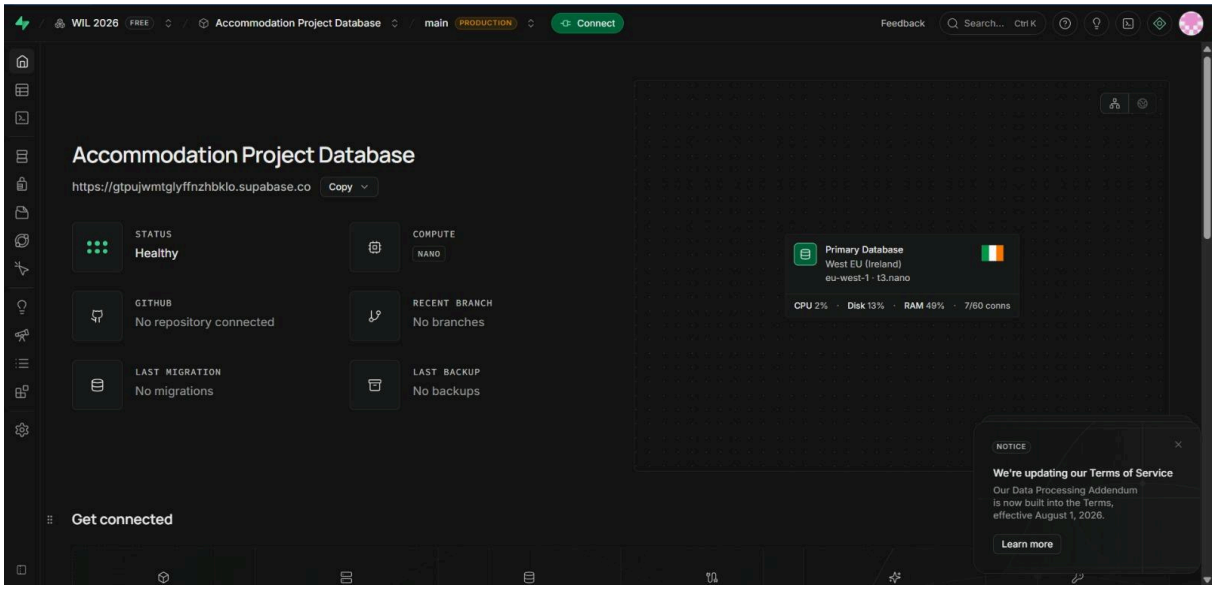
https://github.com/NhlanhlayethuMazibuko/ITC327W-Group-Project.git

15.3 Team Members/Collaborators:



16. SUPABASE EVIDENCE

Supabase Project dashboard:



17. GROUP REFLECTION

Most important learning from this phase:

The group learned the importance of understanding the stakeholder's actual problem and current processes before designing a software solution.

What went well and why:

The group successfully identified Big Brother Accommodation as the stakeholder, conducted stakeholder engagement, gathered requirements and identified the main challenges with the current manual accommodation process.

What was difficult and why:

Finding a suitable external stakeholder was initially difficult. The stakeholder also took some time to respond to requests, which delayed requirements gathering and stakeholder engagement activities.

How were tasks divided among members?:

Tasks were divided according to assigned responsibilities, including stakeholder engagement, requirements, feasibility and risk analysis, documentation, scheduling, GitHub/Supabase setup and progress tracking.

Did all members contribute fairly? Explain:

Yes. Members contributed fairly by respecting their assigned roles and providing the work and deliverables they were expected to complete.

How did the group handle disagreements or delays?:

The group communicated through the WhatsApp group, followed up on outstanding tasks and adjusted responsibilities or timelines where necessary. Stakeholder-response delays were managed through follow-ups and by using the information already provided by the stakeholder.

What did the group learn from the stakeholder?:

The group learned that the accommodation business currently relies heavily on physical viewing, word-of-mouth and WhatsApp-based advertising, while bookings and payments are largely handled manually. The stakeholder also highlighted the need for easier access to accommodation information, applications and bookings, better management of student information and protection against accommodation scams.

Improvement action 1 for the next phase:

Ensure that every member records completed work, dates and supporting evidence in the progress tracker as the work is completed.

Improvement action 2 for the next phase:

Confirm and review requirements with the stakeholder before moving into design and development to ensure that the proposed system remains aligned with the stakeholder's actual needs.

Priority for the next phase:

Finalise and validate the requirements and project scope, then prepare the system design based on the stakeholder's confirmed needs.